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Bayesian Techniques in Macroeconomics

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Master in Quantitative Economics, second year

Gauthier Vermandel[1] *gauthier.vermandel@dauphine.psl.eu*

Chapter 1

Modern Business Cycle Theory

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Handout Objectives

- Understanding the concept of a business cycle.
- Extracting the business cycle component from time series.
- Solving a simple rational-expectations model.
- Linearizing a system of equations.

[1]Department of Economics, Paris-Dauphine University. Updated slides and codes available on my website: https://vermandel.org/btm/.

Session roadmap

Economic intuition. Business cycles combine shocks with propagation mechanisms. **Econometric intuition.** Detrending determines which frequencies the model must explain. **Model and data.** We map quarterly GDP into a cyclical component and then into a linear expectations system. **MATLAB/Dynare exercise.** Reproduce the filters, solve the toy model, and interpret impulse responses. **Takeaway.** Bayesian analysis begins with an explicit frequency convention and transparent model–data mapping.

1. What is a business cycle?

1.1 A brief history

The development of national accounting, especially from the 1930s onward, gave economists more systematic measures of aggregate production from the expenditure, production and income sides. National accounts provide a complete and consistent set of techniques to measure a country's economic activity. Business-cycle research predates modern national accounting, but these measures expanded its empirical foundations. As explained by Schumpeter [1930], the pioneer of the concept of business cycles can be traced back to Mitchell [1913]'s work on the interpretation of gold, prices, and wages during the US Civil War in his treatise on *Business Cycles*. He provided the first quantitative study of business cycles. His pioneering work laid the foundation for modern business cycle analysis in Burns and Mitchell [1946] by describing a business cycle as (punctuation adapted) "a *type of fluctuation found in the aggregate economic activity of nations that organize their work mainly in business enterprises: a cycle consists of expansions occurring at about the same time in many economic activities, followed by similarly general recessions, contractions, and revivals which merge into the expansion phase of the next cycle. In duration, business cycles vary from more than one year to ten or twelve years. They are not divisible into shorter cycles of similar characteristics with amplitudes approximating their own.*"

The contributions of Mitchell helped organize the distinction between long-term growth and short-term fluctuations in macroeconomics. In particular, Burns and Mitchell [1946] provided the first definition of the duration of a business cycle, which lies between one and twelve years. This definition of the business cycle has evolved over time. Modern business cycle analysis determines the frequency of a business cycle as a fluctuation lying between 6 and 32 quarters, which is shorter than the duration suggested by Burns and Mitchell [1946]. Components with periods between 6 and 32 quarters are retained as business-cycle fluctuations. Components with periods longer than 32 quarters are assigned to the trend.

A clear typology of an economic cycle is a primary aspect of macroeconomic analysis, particularly for policymakers. Consider a government facing an adverse event (financial turmoil, pandemic, war, etc.). Dating the recession, gauging its magnitude and duration are all crucial for determining the size of any fiscal/monetary stimulus to mitigate the crisis.[2] During the last century, economic associations, exemplified by the NBER and CEPR, were established to provide start and end dates of recessions for the US and Europe, respectively. Therefore, significant efforts have been made by economists to clearly typify the economic cycle by disentangling periods of expansion, crisis, recession, and recovery in macroeconomic time series.

[2]For instance, in the European Union, there has been ongoing debate since the Maastricht Treaty about fiscal rules and public deficits. In order to maintain a balanced government budget, the EU may penalize any member state exhibiting excessive structural deficits. The critical aspect lies in the definition of the 'structural' nature of the deficit, which typically relies on business cycle measures. A change in the computation of the cyclical budget deficit may alter the stance of the EU Commission on national fiscal rules.

Prior to the 1940s, much of the economic cycle analysis focused on determining the periodicity of cycles. You may have heard during your undergraduate studies about the Juglar, Kuznets, and Kondratiev cycles. These cycle analyses aim to explain how long an innovation, investment, or other spending may create a long-lasting cycle (i.e., determining the duration of an expansion cycle followed by a recession). Following the contributions of Burns and Mitchell [1946], cycle analysis shifted towards disentangling time series into cyclical versus trend components. This change in cycle analysis also resulted in some theoretical interpretations of the cycle through the establishment of short-run and long-run macroeconomics (fluctuations vs. growth). These materialize through new theories of economic growth (e.g., Harrod [1948], Solow [1956]) and theories of fluctuations (e.g., Keynes [1937], Samuelson [1948]). In this lecture, we will focus solely on the business cycle theory which emerged in the 1970s and 1980s.

1.2 Measuring business cycles

However, the frontier separating short-run and long-run analysis is not as clear as one would expect. The emergence of econometrics in the second half of the 20th century had also been accompanied by new statistical methods that aim at separating low- and high-frequency components in economic time series. Any macroeconomist working with time series faces the following problem:

$$y_t = \tau_t + \psi_t, \tag{1}$$

where τ_t is the secular and ψ_t the cyclical component of time series y_t. How can we compute τ_t and ψ_t? Unfortunately, there is no unique method to disentangle τ_t and ψ_t. Therefore, the measure of business cycles is still an open question. In this section, we will discuss three different methods to extract the business cycle component from the data.

A preliminary step before working on real data is to apply logs. Why do we use logs? Because GDP data exhibit an exponential shape. To illustrate this aspect, let us assume for now that economic growth is following an exponential growth process (a very common assumption). Formally, let Y_t denote the GDP that follows the following data-generating process:

$$Y_t = Y_0 \left(1+\gamma\right)^t \exp(\varepsilon_t). \tag{2}$$

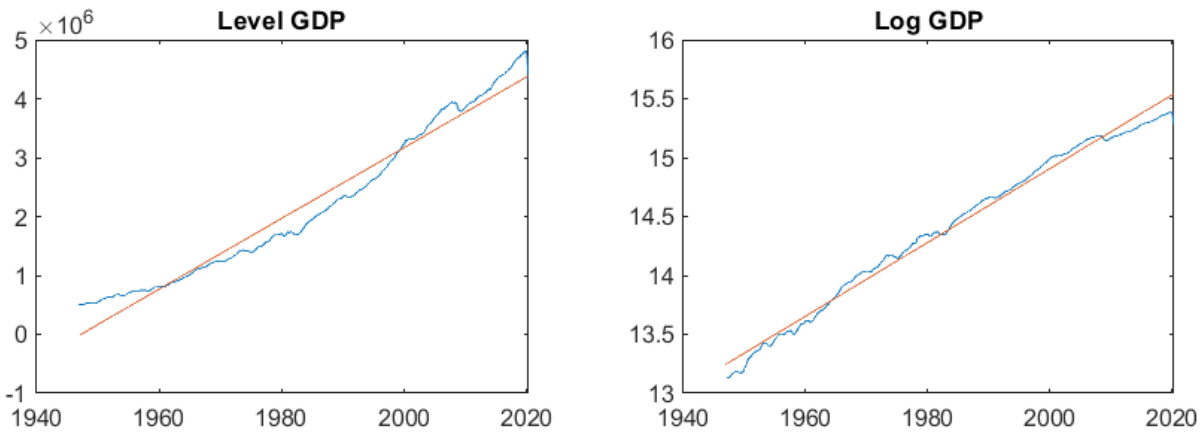
In this expression, $(1+\gamma)^t$ is the (exponential) trend, ε_t is the fluctuation term and Y_0 is the initial trend level (observed initial output also contains the disturbance). In this expression, the interpretation of γ is simply the quarterly growth rate in decimal units. Applying logs to this expression, we 'linearize' the expression (2) by flattening the exponential shape of output:

$$\log(Y_t) = \log(Y_0) + \log(1+\gamma)t + \varepsilon_t \tag{3}$$

$$\simeq \underbrace{y_0 + \gamma \cdot t}_{\tau_t} + \underbrace{\varepsilon_t}_{\psi_t}. \tag{4}$$

OLS estimates the intercept y_0 and exact log-growth slope $g = \log(1+\gamma)$. Recover the level growth rate as $\gamma = \exp(g) - 1$. For small growth, $g \simeq \gamma$. From these estimated parameters, the trend can be computed and removed from log output to obtain what we usually interpret as the fluctuation term ψ_t in the GDP time series. See the first MATLAB box to see alternative filtering methods to extract ψ_t from Y_t.

MATLAB box 1 logGDP.m



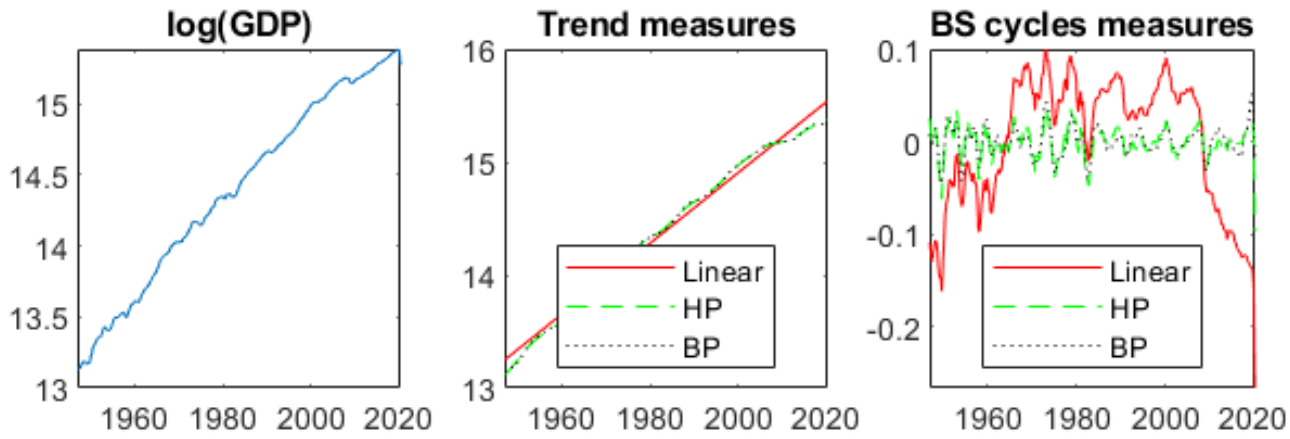
We apply three filtering methods on the log of GDP to illustrate some alternative measures of business cycles:

1. A linear trend: In a nutshell, we use OLS regression to estimate the slope of growth. This method assumes that the data-generating process is the same as in Equation 2. It gives a simple cyclical residual, conditional on a constant log-growth trend. This restriction can discard relevant changes in the growth process. However, if output permanently shifts following a structural break, then the measure of the business cycle may be biased. For example, following the Great Recession in 2007, the US never returned to the pre-crisis trend, resulting in a permanently negative business cycle component after 2007. Considering the potential for structural changes in economic growth, it may be useful to consider a time-varying trend.

2. The HP filter: The statistical method of Hodrick and Prescott [1997] extracts the cyclical component of a time series by separating long-term and short-term fluctuations. This method is two-sided, meaning that for computation in period t, it requires both past and future values. Thus, endpoint trend estimates are particularly sensitive to future data revisions and the treatment of sample boundaries. In addition, the filter may omit information in the data that could be relevant for analysis. This method has faced many critiques, summarized by Hamilton [2018].

3. The band-pass filter: A band-pass filter is a device that passes frequencies within a certain range and attenuates frequencies outside that range. In economics, the band-pass filter can be used to isolate the component of a time series within a particular frequency band to separate the business cycle, the medium-term cycle, and the long-term cycle (i.e., the trend). This is useful, for instance, to examine if an economic relationship (e.g., money-inflation) remains significant across different frequencies of the cycle. Baxter and King [1999] and Christiano and Fitzgerald [2003] provide the two most renowned applications of the band-pass filter in business cycle analysis.

MATLAB box 2 BSmeasures.m



The MATLAB-box figures are historical illustrations, not new observations. The scripts use an archived OECD/QNA vintage. The band-pass output retains the 6–32-quarter component. Its complement contains slower and faster movements. The historical label 'trend' for that complement is not a pure low-frequency trend. The HP choice $\lambda = 1600$ solves

$$\min_{\{\tau_t\}} \sum_{t=1}^{T} (y_t - \tau_t)^2 + \lambda \sum_{t=2}^{T-1} \Big[(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1}) \Big]^2.$$

The first term penalizes deviations from the data, the second changes in trend growth. Neither this choice nor a 6–32-quarter band uniquely defines a business cycle.

1.3 The analysis of business cycles

Blanchard and Watson [1986] provides a nice description of modern business cycle analysis: "*Fluctuations in economic activity are seen as the result of small, white noise shocks that affect the economy through a complex dynamic propagation system.*" Pioneered by Frisch [1933], theoretical interpretations of economic fluctuations (i.e. the term ε_t in Equation 2) rely on system response analysis following the realization of some shocks. The dominant view is that even large fluctuations result from small shocks.[3] Particular economic fluctuations can be ascribed to particularly large shocks followed by periods during which the economy returns to equilibrium. This interpretation contrasts with the "old" Keynesian approach of economic fluctuations: the latter interpret fluctuations as imbalances in money or labor markets.

According to Lucas Jr [1977], the business cycle exhibits some regularities stable over time:

1. The main regularity is at a sectoral level: output movements across sectors are positively correlated. If one sector experiences a boom, other sectors often experience a boom too.
2. Production of durable goods is more volatile than non-durable goods.
3. Prices are procyclical in the historical regularities discussed by Lucas.
4. Short term interest rates are procyclical.

[3] Some more recent applications to financial turmoils, such as Kiyotaki and Moore [1997] or Boissay et al. [2016], show that small shocks can trigger large depressions.

5. Monetary aggregates and velocity measures are procyclical.

Given these observations, Lucas Jr [1977] argued that "*all business cycles are alike*". This assertion motivates a search for common mechanisms behind fluctuations. It does not establish that one particular real business cycle model explains every regularity. Real business cycle theory is one influential development of this research programme. In the following years, Lucas provided several critical contributions to the foundation of a theory of business cycles. The main ingredients are a general equilibrium with infinitely-living and representative agents that form their expectations rationally. This handout provides the basis of business cycle theory pioneered by Bob Lucas.

2. An illustration of a business cycle model through the consumption-saving problem

According to Lucas Jr [1977], a theory of business cycles aims at explaining the value of output at any point in time. In particular, it provides some theoretical interpretations of the empirical relations across macroeconomic time series, based on agent-level decisions (usually referred to as *microfoundations*). In this section, we present the consumption-saving problem to examine how changes in (expected) consumption affect the equilibrium real rate.

2.1 A toy model of consumption-saving

Consider a simplified version of Lucas Jr [1978]'s asset pricing model. This model is characterized by discrete-time and an infinite horizon economy populated by *households*, which are infinitely lived and of measure one. This is an endowment economy, so the endowment is assumed to be fully exogenous and driven by an exogenous shock:

$$y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t \text{ with } \varepsilon_t \sim \mathcal{N}(0, \sigma^2), \tag{5}$$

The production in this economy stems from the presence of infinitely lived trees, output is the fruit that falls from trees. Therefore, we interpret ε_t as the weather that disrupts the quantity of fruits that the tree can provide (i.e., for a low realization of ε_t, the tree provides a lower quantity of fruits). The weather exhibits some persistence, so a low realization of the weather today is likely to affect the quantity of fruits in the next period by a degree ρ. To illustrate this persistence, a storm may damage a branch, the latter requires few periods to heal and produce fruit as usual. In this model, the only source of uncertainty is about current and future values of ε_t, as the household does not know how much fruit she will have. This uncertainty can be addressed by the purchase of some financial claims that we discuss hereafter.

Regarding the preferences, each household maximizes her lifetime utility, measured as the current and future discounted streams of utility:

$$\max_{\{c_{t+s}, b_{t+s}\}_{s\geq 0}} \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s u\left(c_{t+s}\right), \tag{6}$$

where $0 < \beta < 1$ is the discount factor. A lower discount factor assigns less weight to future utility relative to current utility. This concerns the weights, not the sign or absolute level of utility. We impose a logarithm for the utility function $u\left(c_t\right) = \log(c_t)$. The operator $\mathbb{E}_t$ denotes expectation conditional on

information available at date t, including the current endowment. Rational expectations use the model's conditional distribution. They do not imply knowledge of future innovations or perfect foresight.

Regarding the financial aspect of the model, we depart from Lucas Jr [1978] to introduce a (risk free) financial claim in the model. A household can transfer purchasing power across dates by buying a one-period risk-free bond. A competitive financial intermediary can interpret this claim, but it does not create aggregate fruit storage or outside funds. A bond pays the gross real return r_t at date $t+1$ per unit invested at t. It is not a complete hedge against aggregate weather risk. Fruit is the numeraire.

So each period, the household has the possibility to consume c_t units of fruits or deposit b_t to the bank. The budget constraint faced by the household at any point in time is given by:

$$c_t + b_t \leq b_{t-1} r_{t-1} + y_t, \tag{7}$$

where the left-hand side of the equation gathers the spending of the household, while the right-hand side gathers the income from deposits and the production from trees. When the household chooses deposits b_t at date t, she knows the gross real interest rate r_t paid at date $t+1$. The net real interest rate is $r_t - 1$.

The Lagrangian problem solved by the household is given by:

$$\mathcal{L}_t = \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s \left[\log\left(c_{t+s}\right) - \lambda_{t+s} \left(c_{t+s} + b_{t+s} - b_{t-1+s} r_{t-1+s} - y_{t+s} \right) \right], \tag{8}$$

where λ_t denotes the Lagrangian multiplier over the budget constraint. Discounting the constraints is a normalization of the multipliers. The budget constraint binds because utility is increasing in consumption. The Euler equation is derived for an interior choice of bonds, together with an appropriate no-Ponzi condition and transversality condition.

Maximizing Equation 8 with respect to c_t and b_t yields the following first-order conditions:

$$(c_t) : \lambda_t = \frac{1}{c_t} \tag{9}$$

$$(b_t) : \lambda_t = \beta E_t \left\{ \lambda_{t+1} \right\} r_t \tag{10}$$

Together with feasibility, admissibility and transversality, these conditions characterize an interior optimum in the concave household problem. We provide an interpretation of these two equations in the following subsection.

2.2 A discussion of the first order conditions

The two first-order conditions (9-10) are one of the main ingredients of real business cycle theory: the Euler equation.

- Marginal utility of consumption (MUC).

In equilibrium, the Lagrangian multiplier over the budget constraint is equal to the marginal utility of consumption: $\lambda_t = u'(c_t)$. The functional form of the utility function $u(c_t)$ critically matters to explain the saving-consumption nexus. So this section provides some insights about the interpretation of the utility function.

Consider an exogenous consumption cycle as defined in Equation 5, what are the implications of this cycle in terms of utility function $u(c_t)$ and marginal utility of consumption λ_t for the household? Figure 1 reports some illustrative simulations of consumption, and their corresponding streams of utility and marginal utility.

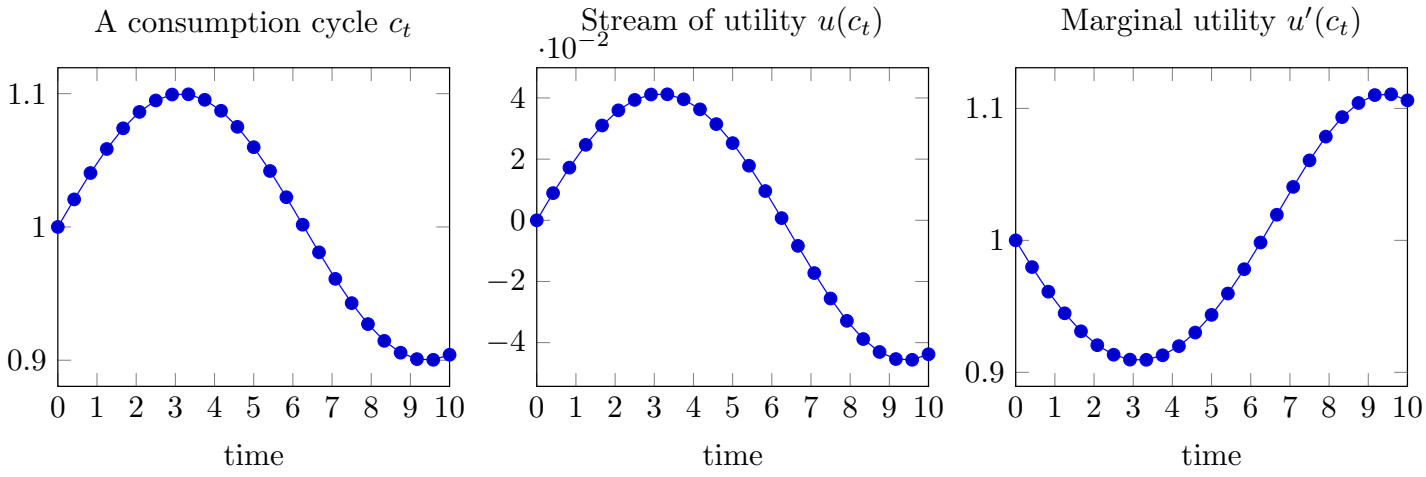


Figure 1: Consumption, utility and marginal utility cycles under logarithmic utility

When consumption is high in period 3, the individual utility also increases with the level of consumption, so utility moves monotonically with consumption. This need not imply a correlation coefficient equal to one. In contrast, the marginal utility exhibits a different pattern of fluctuations with respect to consumption. When consumption is high, the willingness to consume is low. Why? Because of **diminishing marginal utility**, the consumption of each additional unit of a good gives less extra utility than the consumption of the previous unit. Think about a cyclist running the Tour de France who is thirsty following some physical effort to run up a hill. At first, he's really willing to drink water (λ_t is high), so the first glass of water provides a lot of utility. For the second glass, he becomes less thirsty so the consumption of the second glass does provide utility but not as much as the previous one. This feature of the marginal utility explains the negative relation between consumption and the willingness to consume. When consumption is low, the utility from consuming an additional unit of goods is higher than when consumption is high.

○ Intertemporal Marginal Rate of Substitution (IMRS).

The second core key ingredient of modern business cycle theory is the Euler equation. Consider optimal condition on bonds:

$$u'(c_t) = \beta E_t \left\{ u'(c_{t+1}) \right\} r_t. \tag{11}$$

The left hand side of this equation provides the utility loss of saving today (priced in terms of present marginal utility), and the right hand side is the utility gain at the next period. As long as this equation holds to equality, there is a no-arbitrage condition that makes the household equally well off between consuming today or tomorrow. Note that the equilibrium interest rate plays here a key role as it directly determines how households are weighting consumption today versus tomorrow: holding lifetime resources fixed, a higher return creates a substitution incentive to postpone consumption. The total response can include wealth effects. Therefore, the key ingredient here is the pricing scheme that affects the trade-off between consuming now and tomorrow. The intertemporal marginal rate of substitution in the deterministic two-period diagram is $u'(c_t)/[\beta u'(c_{t+1})]$. Under uncertainty the stochastic discount factor is $\beta u'(c_{t+1})/u'(c_t)$, and its conditional mean equals $1/r_t$.

The intertemporal analysis of consumption is actually exactly the same as in the standard consumer problem describing the maximization of the utility by determining the optimal quantity of two types of goods under a budget constraint. The difference here is that instead of having two different goods, the trade-off now applies between consuming today and tomorrow. The budget constraint becomes intertemporal and shapes the dynamics of the interest rate.

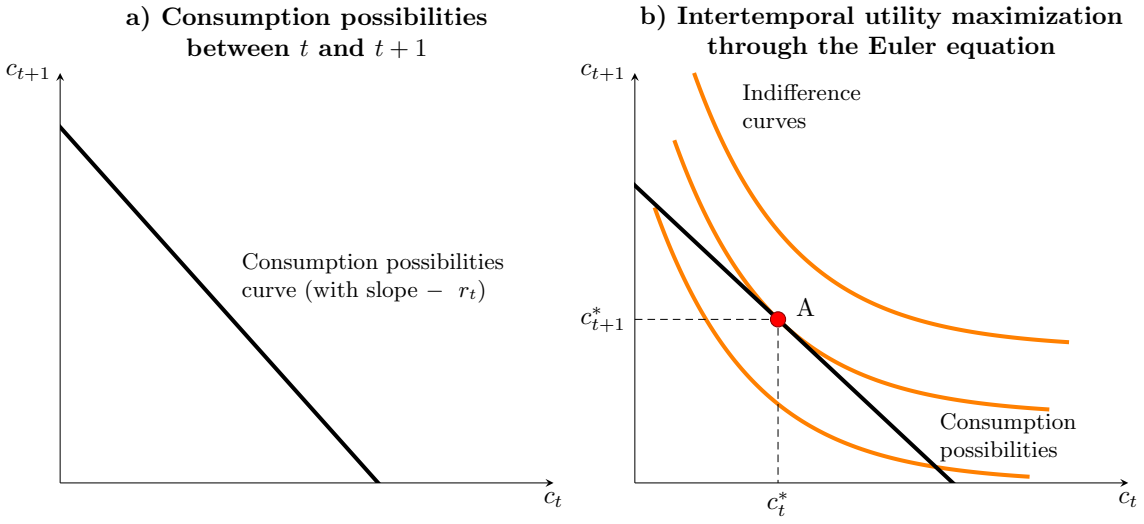


Figure 2: Optimization scheme for the representative household

Figure 2 reports the joint-determination of the optimal level of financial assets and consumption. In the left panel, we show the consumption possibilities (c_t, c_{t+1}) which are offered by financial markets in the economy (and implicitly here by the production capacities of the economy which are exogenous). The slope of this budget line is $-r_t$. Saving one unit of current consumption provides r_t units of future consumption.

In the right panel of Figure 2, we introduce the household's preference with respect to consumption. These preferences materialize in a graphical representation as indifference curves. These curves are different pairs (c_t, c_{t+1}) that provides exactly the same utility so that an individual is equally well off at any point of the indifference curve. Given the real rate on deposits, the household determines the levels of present and future consumption (c^*_t, c^*_{t+1}) maximizing her utility given the consumption possibilities offered by financial markets. In equilibrium A, the household finds the unique possible path of consumption maximizing her utility intertemporally and is called *the consumption point*. The tangency is the Euler condition in this deterministic two-date illustration. It must hold together with the budget constraint.

2.3 General Equilibrium

Finally, we need to set one condition over deposits to close the model. In financial autarky (*i.e.*, with no trading partners), the general equilibrium is jointly determined by the equilibrium on financial and goods markets, as the optimal condition of saving determine the amount of consumption. So clearing one market also clears the others.

In absence of external borrowing/lending, at a macroeconomic level, the following condition on domestic saving emerges:

$$b_t = 0. \tag{12}$$

Note that since households populating our economy are all symmetric, the agent specific saving is also zero in equilibrium. Given this relation, the budget constraint in Equation 7 becomes:

$$c_t = y_t. \tag{13}$$

Therefore, the model can be cast as the following set of two equations:

$$\begin{cases} y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t \\ r_t = \left[\beta y_t E_t\{y_{t+1}^{-1}\}\right]^{-1} \end{cases}. \tag{14}$$

The first equation is the quantity of fruit falling from trees, and the second equation is the equilibrium condition on deposits. In absence of endogenous production, the number of fruit over time determines the equilibrium interest rate that ensures that Equation 12 holds to equality.

Households still choose consumption and bonds optimally, taking prices as given. Market clearing forces aggregate consumption to equal the endowment. Zero equilibrium bond holdings are an outcome, not a ban on saving choices. So the interpretation of the model is the following: the agent in t contemplates how much fruit y_t she has by observing ε_t (i.e. the weather). Given this new information, the agent plans how much fruits y_{t+1} she may expect at the next period. Given the current and expected levels of consumption, the equilibrium rate adjusts in order to clear financial markets, i.e. the real rate ensures that saving remains zero in equilibrium.

2.4 Expectations

How do agents form the conditional expectation of future marginal utility in system (14)? To illustrate the forecasting step, we first compute $E_t\{y_{t+1}\}$. Replacing future output by this conditional mean gives a certainty-equivalent approximation, not the exact stochastic Euler equation. Expectation formation remains an important topic in macroeconomics.

Many large-scale macroeconometric models of the 1960s relied heavily on backward-looking behavioural equations. The Lucas critique Lucas et al. [1976] stresses that policy changes can alter the expectations and decision rules underlying estimated relationships. Adding an expectation term alone does not resolve that policy-invariance problem.

Which forward looking dynamics are we talking about? Let us think about some simple examples in macroeconomics:

1. A first example is about **price setting**. If the central bank is accommodative today, a worker may expect that inflation will rise in the future, so the worker asks today for an increase of her wage to compensate future inflation (which in turn self-fulfills the increase in inflation now). These considerations led to a new interpretation of the Keynesian Phillips curve.

2. Another example concerns **portfolio allocation in open economies**. Suppose that a central bank in the domestic country raises its interest rate. Holding the expected future exchange rate and the foreign rate fixed, uncovered interest parity implies an immediate appreciation of the domestic currency followed by expected depreciation. The expected exchange-rate loss offsets the higher domestic interest return. These two examples are simply not incorporated in purely backward models.

The rational-expectations hypothesis, pioneered by Muth [1961], requires forecasts consistent with the model and the information available to agents. This restriction does not make every model easy to solve. Here, the endowment law is known and innovations are independent with conditional mean zero. So if our agent knows the endowment in Equation 5, by iterating forward this equation we get:

$$\begin{aligned} E_t\left\{y_{t+1}\right\} &= E_t\left\{1-\rho+\rho y_t+\varepsilon_{t+1}\right\} \\ &= E_t\left\{1-\rho+\rho y_t\right\} \\ &= 1-\rho+\rho y_t, \end{aligned}$$

So as you can see, we are able to transform $E_t\left\{y_{t+1}\right\}$ into a system of current and backward variables. Implicitly, you may have noticed that we imposed $E_t\left\{\varepsilon_{t+1}\right\}=0$. This is because the agent ignores the

future realizations of the shock, but knows its distribution. Here independence implies $E_t(\varepsilon_{t+1}) = 0$. An unconditional zero mean alone would not justify that conditional forecast. So agent's expectations may be wrong, but **are correct on average** over time.

Forecast errors need not vanish in any given period:

$$y_{t+1} - E_t\{y_{t+1}\} = \varepsilon_{t+1}.$$

Under rational expectations they have conditional mean zero:

$$E_t\left[y_{t+1} - E_t\{y_{t+1}\}\right] = 0.$$

Under the usual law-of-large-numbers conditions, their sample average converges to zero. Individual forecast errors do not converge to zero.

Back to our two-equation system in Equation 14, under rational expectations and the certainty-equivalent approximation, the model becomes:

$$\begin{cases} y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t \\ r_t \simeq \frac{(1-\rho+\rho y_t)}{\beta y_t} \end{cases}. \tag{15}$$

For positive simulated endowment, this certainty-equivalent system gives the associated gross real rate.

Exercise 1: Simulating the certainty-equivalent model

Consider the system in Equation 15.

1. Write a MATLAB file that simulates this system over 400 periods. Use the following calibration: $\beta = 0.99$, $\sigma = 0.1$ and $\rho = 0.85$.
   (a) Set the following calibration: $\beta = 0.99$, $\sigma = 0.1$ and $\rho = 0.85$.
   (b) Compute the steady state of the model.
   (c) Draw 400 exogenous realizations of ε_t from a Gaussian distribution using function `randn()`.
   (d) Declare an empty vector using function `nan()` for `y` and `r` that we will fill next.
   (e) Run a loop that fills recursively the vector for `y` and `r`.
2. Plot the simulation.
3. Compute the standard deviation of output and its first-order autocorrelation. Compare $\log y_t$ (locally $y_t - 1$) with the log-GDP residual from the linear trend, using the same units. Do not compare output in model levels with a percentage GDP cycle.
4. If you find a gap, adjust the calibration of σ and ρ to match US data.
5. Simulate an Impulse Response Function (IRF) of the model.
6. Does a large output expansion have the same effect on the real rate as a large depression? Compare the absolute values of the IRFs.
7. Plot the histograms of your simulated data and explain why the asymmetry is different for the distribution of output and the real rate.

2.5 A discussion on stochastic uncertainty

The solution in Equation 15 ignores risk corrections. Even with log utility, $E_t\{y_{t+1}^{-1}\}$ is not generally equal to $1/E_t\{y_{t+1}\}$. With CRRA utility and curvature $\sigma_C > 0$,

$$u'(c) = c^{-\sigma_C}, \qquad r_t = \frac{y_t^{-\sigma_C}}{\beta E_t\{y_{t+1}^{-\sigma_C}\}}.$$

Here σ_C is distinct from the innovation standard deviation σ. The certainty-equivalent expression replaces future output by its conditional mean. It gives the correct first-order approximation around the deterministic steady state, but not generally higher-order risk effects. The additive Gaussian endowment is only a local teaching approximation. It assigns positive probability to non-positive consumption, and its inverse has no ordinary finite expectation. Thus the exact Euler expression is a formal benchmark requiring positive support and finite expected marginal utility, not a globally well-defined Gaussian economy. Positivity in a finite sample does not repair that distribution. The file named toy_model_bootstrap.m uses conditional Monte Carlo draws rather than resampling observations. It does not compute an exact risk correction.

The Dynare variant uses $y_{t+1} = y_t^{\rho}\exp(\varepsilon_{t+1})$, with innovation variance σ^2 and curvature σ_C. It has positive support and finite inverse moments. The Gaussian exponential moment gives

$$E_t[y_{t+1}^{-\sigma_C}] = y_t^{-\rho\sigma_C}\exp(\sigma_C^2\sigma^2/2), \qquad r_t = \beta^{-1}y_t^{\sigma_C(\rho-1)}\exp(-\sigma_C^2\sigma^2/2).$$

This exact expression checks this variant only. The supplied file uses curvature 10 and an order-two approximation. Set curvature to one and choose order one explicitly when comparing the log-utility local rules.

3. Approximation methods

In the two-dimensional model considered, it is possible to simulate the certainty-equivalent non-linear system explicitly. However, if the model gets more complicated (in particular with state variables), it's not possible to get a closed-form expression of the non-linear model. Therefore, approximation methods are applied to the model's equation to obtain a closed-form expression of a linear transition equation.

3.1 The problem

Our 2 equation model is a non-linear system with rational expectations. In compact notation, the system can be represented as follows:

$$\mathbb{E}_t\left[f_\theta\left(x_{t+1}, x_t, x_{t-1}, \epsilon_t\right)\right] = 0 \tag{16}$$

where:

- x is the vector collecting n endogenous variables.
- ϵ is the vector collecting the p exogenous stochastic shocks.
- θ is the vector collecting the deep parameters of the model.
- $f_\theta(.)$ is the equations of the model.

The solution to a model like this is normally a set of functions relating controls to states such that our 2 equations are satisfied at all times (i.e. in t+1, t and t-1). The solution of the system x_t takes the form a decision rule:

$$x_t = g\left(x_{t-1}, \epsilon_t\right), \tag{17}$$

where function $g(.)$ is unknown, but theoretically can be computed from Equation 16. Let us first rewrite the system in Equation 16 by injecting the model's solution in Equation 17. Under rational expectations, $E_t\{x_{t+1}\} = E_t\{g(x_t, \epsilon_{t+1})\}$. The exact equilibrium condition retains this expectation:

$$E_t\left[f_\theta\left(g(x_t, \epsilon_{t+1}), x_t, x_{t-1}, \epsilon_t\right)\right] = 0. \tag{18}$$

Iterating again to express our system as a function of x_{t-1}:

$$E_t\left[f_\theta\left(g(g(x_{t-1}, \epsilon_t), \epsilon_{t+1}), g(x_{t-1}, \epsilon_t), x_{t-1}, \epsilon_t\right)\right] = 0 \tag{19}$$

Therefore, the true solution of the system Equation 16 is a function $g(.)$ that satisfies Equation 19. The problem is that, in case of larger models, there is an infinite number of values of x_{t-1} and ϵ_t and an infinite number of values of x_t to find. Hence, except for few exceptions, the usual practice is to approximate $g(.)$.

Note that in our two-dimensional model of consumption-saving, we have a closed-form certainty-equivalent rule $g(.)$ but in more complicated models, it's generally necessary to use numerical solution methods. The next subsection aims at explaining how we can approximate $g(.)$.

The closed-form certainty-equivalent policy rule $x_t = g\left(x_{t-1}, \epsilon_t\right)$ is given by:

$$y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t \tag{20}$$

$$r_t = \frac{\rho}{\beta} + \frac{\beta^{-1}\left(1-\rho\right)}{1 - \rho + \rho y_{t-1} + \varepsilon_t}. \tag{21}$$

3.2 Linearization Method

The standard way to bound the problem is to approximate the function $g\left(.\right)$ by a log-linear function around the steady state. There are many different solution methods (e.g. Schur decomposition, undetermined coefficients, etc) to reach the linearized version of $g\left(.\right)$. Under particular conditions defined by Blanchard and Kahn [1980], our linear rational expectations model has a stable and convergent intertemporal equilibrium.

More precisely, we usually take the following steps to solve a model:

1. Find the steady state of the model.
2. (log-)Linearize around the steady state.
3. Write the Dynare file.

For any function $f(x_t)$ with corresponding steady state $f(\bar{x})$, a Taylor expansion up to first order is given by:

$$f(x_t) \simeq f(\bar{x}) + f'(\bar{x})\left(x_t - \bar{x}\right) \tag{22}$$

Expressed in terms of deviation from the steady state, $f(x_t) - f(\bar{x})$, the expansion can be rewritten in the following way:

$$\begin{aligned} f(x_t) - f(\bar{x}) &\simeq f'(\bar{x})\left(x_t - \bar{x}\right) \\ &\simeq f'(\bar{x})\bar{x}\frac{(x_t - \bar{x})}{\bar{x}} \\ &\simeq f'(\bar{x})\bar{x}\log(\frac{x_t}{\bar{x}}) \\ &\simeq f'(\bar{x})\bar{x}\hat{x}_t \end{aligned}$$

where $\hat{x}_t = \log(x_t/\bar{x})$ is a log deviation in decimal units. Multiplying it by 100 gives an approximate percentage deviation.

For a two-variable function, the approximation reads as:

$$f(x_t, y_t) - f(\bar{x}, \bar{y}) \simeq f'_x(\bar{x}, \bar{y})\bar{x}\hat{x}_t + f'_y(\bar{x}, \bar{y})\bar{y}\hat{y}_t \tag{23}$$

We can then summarize these general approximations with a few rules that we can use to log-linearize quickly a model:

Level variable	First-order deviation from steady state
x_t	$\bar{x}\hat{x}_t$
$x_t y_t$	$\bar{x}\bar{y}(\hat{x}_t + \hat{y}_t)$
$\frac{x_t}{y_t}$	$\frac{\bar{x}}{\bar{y}}(\hat{x}_t - \hat{y}_t)$
x_t^a	$\bar{x}^a a\hat{x}_t$
$\log(x_t)$	$\hat{x}_t$
$\exp(x_t)$	$\exp(\bar{x})\bar{x}\hat{x}_t$

3.3 Linearization application

Let us now linearize our system of equations:

$$\begin{cases} y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t \\ r_t \simeq \frac{(1-\rho+\rho y_t)}{\beta y_t} \end{cases}. \tag{24}$$

Steady state. A preliminary step before performing the approximation to our model's equations is to characterize the steady state of the model (i.e. the fixed point of the approximation). Following the linearization, each positive variable is expressed as a log deviation in decimal units. Multiply by 100 for an approximate percentage deviation. To obtain the steady state, one just needs to take off the time subscript on endogenous variables:

$$\begin{cases} \bar{y} = 1 - \rho + \rho\bar{y} \\ \bar{r} = \frac{(1-\rho+\rho\bar{y})}{\beta\bar{y}} \end{cases}. \tag{25}$$

Note that the following system in the (deterministic) steady state reads as follows:

$$\begin{cases} \bar{y} = 1 \\ \bar{r} = \frac{1}{\beta} \end{cases}. \tag{26}$$

Linearizing the equations. Next, we must linearize the equations of the model. We apply here:

$$y_t - \bar{y} = \rho(y_{t-1} - \bar{y}) + \varepsilon_t, \qquad \bar{y}\hat{y}_t \simeq \rho\bar{y}\hat{y}_{t-1} + \varepsilon_t.$$

Note that since $\bar{y} = 1$, then equation boils down to:

$$\hat{y}_t \simeq \rho\hat{y}_{t-1} + \varepsilon_t$$

Some remarks on the linearization. You may struggle to log-linearize, but with learning-by-doing, you will get used to it. After some time of practice, linearizing equations becomes quicker. Note also that the stochastic shock is kept in levels because its steady state is zero.

$$
\begin{aligned}
r_t &= \frac{(1-\rho+\rho y_t)}{\beta y_t} \\
r_t &= \frac{(1-\rho)}{\beta}\frac{1}{y_t} + \frac{\rho}{\beta} \\
\bar{r}\hat{r}_t &\simeq -\frac{1-\rho}{\beta\bar{y}}\hat{y}_t \\
\hat{r}_t &\simeq -(1-\rho)\,\hat{y}_t
\end{aligned}
$$

So the linearized system reads as:

$$
\begin{aligned}
\hat{y}_t &\simeq \rho\hat{y}_{t-1} + \varepsilon_t \\
\hat{r}_t &\simeq -(1-\rho)\,\hat{y}_t
\end{aligned}
$$

And we are looking for a solution of the form for $\hat{x}_t = [\hat{y}_t, \hat{r}_t]'$:

$$
\hat{x}_t = P\hat{x}_{t-1} + Q\epsilon_t
$$

While equation for $\hat{y}_t$ is recursive, the other in $\hat{r}_t$ is not. We simply inject $\hat{y}_t$ in $\hat{r}_t$ to force recursion: $\hat{r}_t \simeq -(1-\rho)\,\hat{y}_t = -(1-\rho)\,\rho\hat{y}_{t-1} - (1-\rho)\,\varepsilon_t$. The solution of the system reads as:

$$
\underbrace{\begin{bmatrix} \hat{y}_t \\ \hat{r}_t \end{bmatrix}}_{\hat{x}_t} = \underbrace{\begin{bmatrix} \rho & 0 \\ -(1-\rho)\,\rho & 0 \end{bmatrix}}_{P}\underbrace{\begin{bmatrix} \hat{y}_{t-1} \\ \hat{r}_{t-1} \end{bmatrix}}_{\hat{x}_{t-1}} + \underbrace{\begin{bmatrix} 1 \\ -(1-\rho) \end{bmatrix}}_{Q}\varepsilon_t
$$

As long as we know vector ε_t, matrices P and Q, and initial conditions $\hat{y}_0$ and $\hat{r}_0$, we are able to characterize the path of $\{\hat{y}_t, \hat{r}_t\}_{t=1}^{\infty}$.

Exercise 2: Simulating the linear model

Consider now the linear system. This exercise can be coded in the same script file as Exercise 1.

1. Write a MATLAB file that simulates this system over 400 periods.
2. Plot the simulation.
3. Simulate an Impulse Response Function (IRF) of the model. An IRF is just the same simulation as before, but with one shock at the first period and next no shocks. It isolates the pure propagation mechanism.
4. Does a large output expansion have the same effect on the real rate as a large depression? Compare the absolute values of the IRFs.
5. Plot histograms of the linearized variables. Explain why Gaussian innovations generate symmetric stationary distributions in this linear model, unlike the nonlinear certainty-equivalent rate.
6. Compute the mean squared approximation error for the gross real rate, $E\left[(r_t - \bar{r}(1+\hat{r}_t))^2\right]$. Compare both series in levels, using the same innovations.

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